



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
FULL SYLLABUS
29-03-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE : [SCORE-II (ALL STAR BATCH)]

Time : 3 Hours

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **36** pages and that all the **19** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

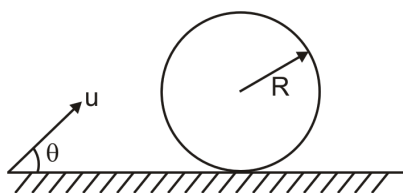
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. A particle is thrown from ground with speed u at angle θ with the horizontal as shown. If u_0 is the minimum value of u so that the particle will cross the horizontal cylinder just grazing at two points on it and θ_0 is the corresponding angle of projection. Then choose the correct statement(s).



(A) $u_0^2 = Rg[2 + 2\sqrt{2}]$

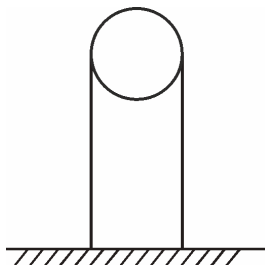
(B) $\tan^2 \theta_0 = [3 + 2\sqrt{3}]$

(C) $u_0^2 = Rg[2 + 2\sqrt{3}]$

(D) $\tan^2 \theta_0 = [3 + 2\sqrt{2}]$

Space for Rough Work

2. There is a long vertical tube of radius r containing air at atmospheric pressure. A steel ball is held at the mouth of the tube and dropped. The ball has radius r and just fits inside the tube. The tube wall is perfectly smooth and no air can leak from the tube as the ball falls inside it. The ball falls through half the length of tube before coming to rest. Assume that wall of the tube is perfectly conducting and temperature of the air inside tube remains constant. Density of steel is d and atmospheric pressure is P_0 and take $L \gg r$, take air to be an ideal gas



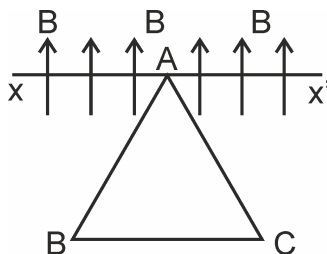
- (A) The radius (r) of the tube is $\frac{3P_0}{2dg} \ln 2$
- (B) The radius (r) of the tube is $\frac{3P_0}{dg} \ln 2$
- (C) Depth from top of the tube, the ball will be in equilibrium is $L/2$
- (D) Depth from the top of the tube, the ball will be in equilibrium is $L \left(1 - \frac{1}{2 \ln 2} \right)$

Space for Rough Work

3. Choose the correct option(s)
- (A) If Q -value of an exothermic nuclear reaction is $+15.76 \text{ MeV}$ and the kinetic energy of products of reaction is 18.24 MeV then kinetic energy of incident particles is 2.48 MeV .
- (B) In a nuclear reaction, during α -decay from a nucleus at rest, energy released is shared between α -particle and daughter nucleus is in the form of kinetic energy and its value for α -particle has more than that of daughter nucleus.
- (C) In a decay process of radio active element, the activity of the sample at any instant is directly proportional to the number of undecayed nuclei present in the sample at that time.
- (D) Photoelectric effect supports quantum nature of light because there is a minimum frequency of light below which no photoelectrons are emitted.

Space for Rough Work

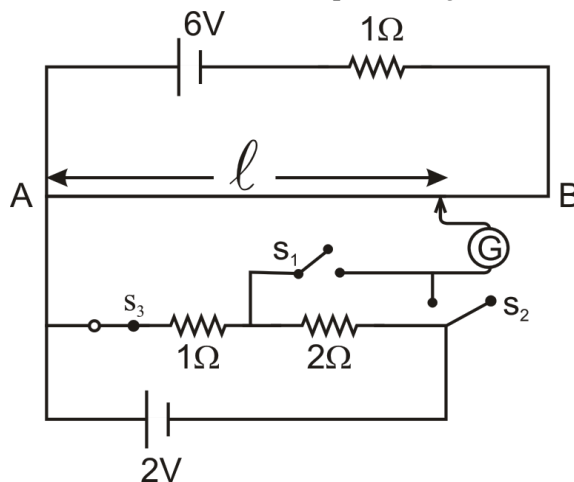
4. A uniform conducting wire of mass $3m$ and length 3ℓ is bent into the shape of an equilateral triangular loop ABC and suspended in a vertical plane from a fixed horizontal axis XX' as shown in the figure. The loop is free to rotate about the fixed horizontal axis XX' . Each side of the triangular loop is having a mass m & length ℓ . A uniform vertical upward magnetic field B is existing in the region. The charge Q is passed through the triangular loop almost instantaneously in the initial vertical position. If the maximum deflection of the triangular loop from vertical plane is $\theta = \pi/2$, then choose the correct option(s).



- (A) The initial angular velocity acquired by the loop is $\sqrt{\frac{2\sqrt{3}g}{5\ell}}$
- (B) The initial angular velocity acquired by the loop is $\sqrt{\frac{8\sqrt{3}g}{5\ell}}$
- (C) The value of magnetic field B is $\frac{m}{Q} \sqrt{\frac{40g}{\sqrt{3}\ell}}$
- (D) The value of magnetic field B is $\frac{m}{Q} \sqrt{\frac{10g}{\sqrt{3}\ell}}$

Space for Rough Work

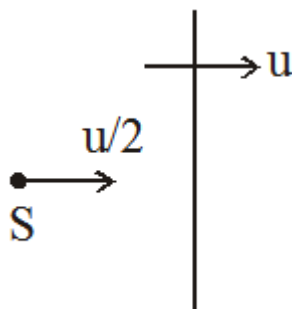
5. Figure shows a potentiometer circuit. The length of potentiometer wire AB is 120 cm and its resistance is 2Ω . The internal resistance of 2V cell is negligible and the internal resistance of 6 V cell is 1Ω , as already shown in the branch. Initially the switches s_1 and s_2 are open and s_3 is closed.



- (A) When only s_2 is open, balanced length $\ell = 20$ cm.
- (B) When only s_1 is open, balanced length $\ell = 90$ cm.
- (C) When all three switches are closed, balanced length $\ell = 60$ cm.
- (D) When s_3 is open, s_2 is also open but s_1 is closed, balanced length $\ell = 40$ cm.

Space for Rough Work

6. A wall is moving with velocity u and a source of sound (frequency f) moves with velocity $\frac{u}{2}$ in the same direction as shown. Assuming that the sound travels with a speed of $10u$, pick the right option or options.



(A) wavelength of incident sound is $\frac{19u}{2f}$.

(B) frequency of the incident sound is $\frac{18}{19}f$.

(C) wavelength of reflected sound is $\frac{u}{f}$.

(D) frequency of the reflected sound is $\frac{12f}{19}$.

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

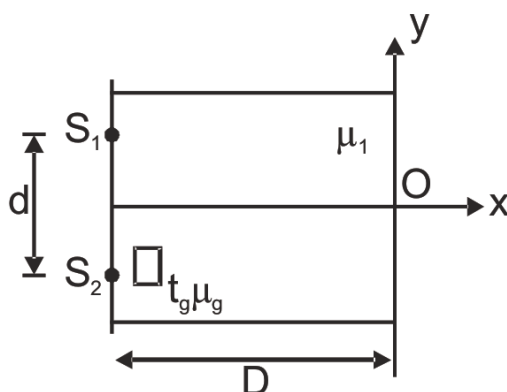
Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

In a modified YDSE, the region between the screen and slits is immersed in a liquid whose refractive index varies with time as $\mu_1 = \frac{5}{2} - \frac{T}{4}$ until it reaches a steady state value of $\frac{5}{4}$. A glass plate of thickness $t_g = 36 \mu\text{m}$ and refractive index $\mu_g = 3/2$ is introduced in front of one of the slits at $T = 0$. (Given $D = 1\text{m}$, $d = 2\text{mm}$.)

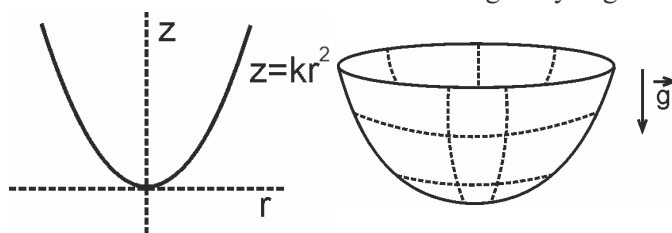


7. The time when central maxima is at O is
 (A) 6 sec (B) 4 sec (C) 3 sec (D) 2 sec
8. The speed of central maxima when it is at O is
 (A) $3 \times 10^{-3} \text{ m/s}$ (B) $6 \times 10^{-3} \text{ m/s}$ (C) $2 \times 10^{-3} \text{ m/s}$ (D) 0

Space for Rough Work

Paragraph for Questions 9 and 10

A particle is constrained to move on the inner surface of a frictionless parabolic bowl whose cross-section has equation $z = kr^2$. The particle begins at a height z_0 above the bottom of the bowl with a horizontal velocity v_0 along the surface of the bowl. The acceleration due to gravity is g .



9. A particular value of horizontal velocity v_0 , which we will name v_h , the particle moves in a horizontal circle is
- (A) $v_h = \sqrt{\frac{2}{3} g z_0}$ (B) $v_h = \sqrt{2 g z_0}$
- (C) $v_h = \sqrt{3 g z_0}$ (D) $v_h = 2 (\sqrt{g z_0})$
10. Suppose that the initial horizontal velocity is now $v_0 > v_h$. What is the maximum height reached by the particle?
- (A) $\frac{v_0^2}{2g}$ (B) $\frac{v_0^2}{4g}$ (C) $\frac{v_0^2}{g}$ (D) $\frac{2}{3} \frac{v_0^2}{g}$

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

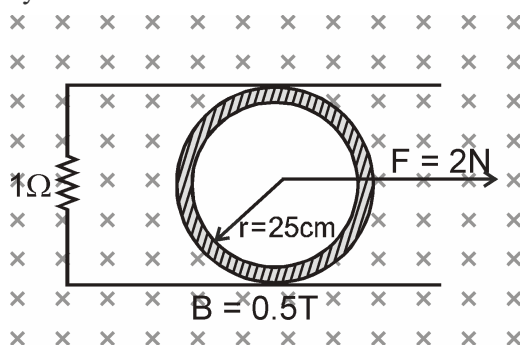
- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

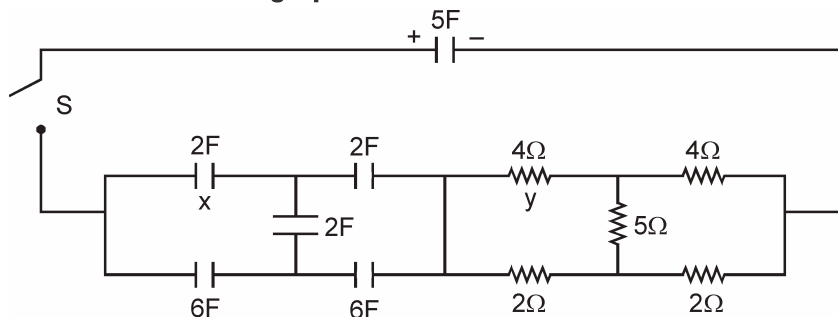
In the shown figure, two parallel rails in a vertical plane of negligible resistance are connected by a resistance $R = 1\Omega$ at one end. A uniform horizontal magnetic field is present ($B = 0.5\text{ T}$). A thin ring of mass $m_1 = 1\text{ kg}$, radius $r = 25\text{ cm}$ is fitted over a wooden disc of mass $m_2 = 1\text{ kg}$. Resistance of ring is $R' = 4\Omega$ and there is no friction between the ring and upper rail. Friction between ring and lower rail is sufficient to support pure rolling of the ring. Horizontal force $F = 2\text{ N}$ is applied at centre of the disc. When the terminal velocity is reached.



1. Terminal velocity of disc in m/s is _____.
2. Current flowing in the rails in A is _____.

Space for Rough Work

Paragraph for Questions 3 and 4

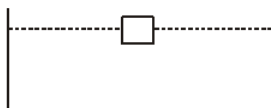


A capacitor of capacity 5F is charged by a battery of 2 volt and then connected with 5 uncharged capacitors and 5 resistors with the help of a switch S as shown in figure. At $t = 0$, switch is closed, then answer the following.

3. Charge on capacitor x (as marked in circuit) at steady state in coulombs is
4. Total energy dissipated in the circuit in joules is

Paragraph for Questions 5 and 6

A cube made of wood having specific gravity 0.4 and side length $a = 1\text{m}$ is floated in a large tank full of water.



5. If the cube is depressed slightly, it executes SHM from it's position. What is it's time period in seconds?
6. What can be the maximum amplitude (in m) of it's vertical simple harmonic motion?

Space for Rough Work

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

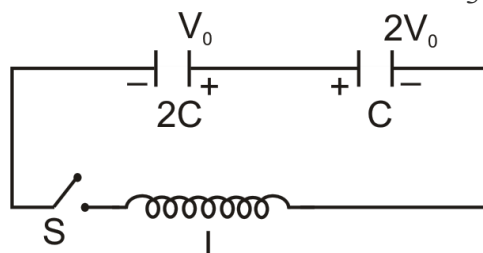
Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

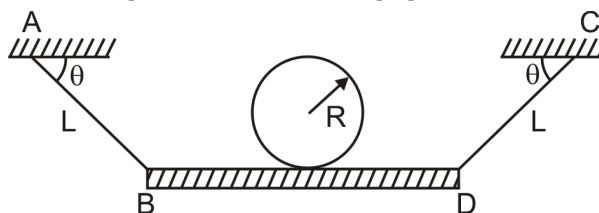
7. The prism spectrum is spread out more at violet end than at the red end. Angular dispersion is defined as $\phi = \frac{d\delta}{d\lambda}$. If deviation of small angled prism δ is defined as $\delta = (\mu - 1) A_0$ and wavelength is defined as $\mu = A + \frac{B}{\lambda^2}$. Where A & B are constants & A_0 is prism angle, then angular dispersion $\phi \propto \frac{1}{\lambda^N}$. Find the value of N.

Space for Rough Work

8. Two ideal charged capacitor $2C$ & C ($C = 2$ Farad) are connected with an ideal inductor of inductance L through a switch S . Initially potential difference across $2C$ is V_0 and across C is $2V_0$, with polarities as shown, where $V_0 = 4$ volts. Also there is no energy stored in the inductor of inductance $L = 3$ Henry. If the maximum current flowing in the circuit after closing the switch is $\frac{n}{3}$ amp. Find the value of n .



9. A metal cylinder of radius R is placed on a wooden plank BD . The plank is kept horizontally suspended with the help of two identical string AB & CD , each of length L . The temperature coefficient of linear expansion of the cylinder and strings are α_1 & α_2 respectively. Angle θ shown in the figure is 30° . It was found that with small change in temperature, the centre of the cylinder did not move. Find $\frac{\alpha_1}{\alpha_2}$. If it is known that $L = 4R$. Assume that change in value of θ is negligible for small temperature changes.



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Which of the following statements are correct ?

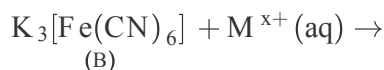
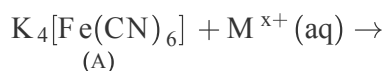
- (A) A plot of Gibb's energy of a reaction mixture at constant pressure and temperature against the extent of reaction shows minima at equilibrium state.
- (B) The melting point of Ice decreases with increase of pressure because Ice contracts on melting.
- (C) Formation of diamond from graphite is favourable at high pressure and high temperature.
- (D) Vapour pressure of a liquid in a closed container is independent of the total surface area of liquid, quantity of liquid and temperature of liquid.

Space for Rough Work

2. Select the incorrect option(s) :-

- (A) Chemisorption is highly specific and it requires activation energy because forces of attraction are vander waal's forces.
- (B) Lyophobic sols are very stable and are not easily coagulated by electrolytes therefore large amount of electrolyte is required to cause its coagulation
- (C) Formation of micelles takes place above kraft temperature and below CMC (Critical micelle Concentration)
- (D) At pH of isoelectric point, the sol particles of amino acids neither move towards anode nor towards cathode because concentration of Conjugate base and Conjugate acid of the zwitter ion of the amino acid become equal.

3. Consider the following reactions



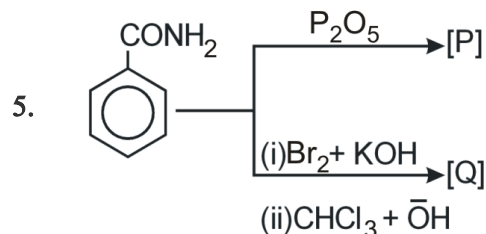
Select the correct set of options in which both give either coloured ppt or white ppt

- (A) $\text{Fe}^{2+}(\text{aq})$ with (A), $\text{Ag}^+(\text{aq})$ with (A) (B) $\text{Fe}^{3+}(\text{aq})$ with (B), $\text{Cu}^{2+}(\text{aq})$ with (B)
- (C) $\text{Zn}^{2+}(\text{aq})$ with (A), $\text{Ag}^+(\text{aq})$ with (B) (D) $\text{Cu}^{2+}(\text{aq})$ with (B), $\text{Fe}^{2+}(\text{aq})$ with (B)

Space for Rough Work

4. Select the incorrect statements among the following options :-

- (A) Fe_2Cl_6 is planar and non-polar where as Fe_2Cl_4 is non-planar and non-polar
 (B) Back bonding is present in H_3SiNCS , GeH_3NCS , $\text{N}(\text{GeH}_3)_3$, $\text{N}(\text{SiH}_3)_3$
 (C) $\text{SiCl}_4 > \text{PCl}_5 > \text{AlCl}_3 > \text{SF}_6$ (Rate of hydrolysis)
 (D) Type of overlapping in the bridge bond is same in $\text{Al}_2(\text{CH}_3)_6$, Al_2Cl_6 , $\text{BeCl}_2(\text{Solid})$

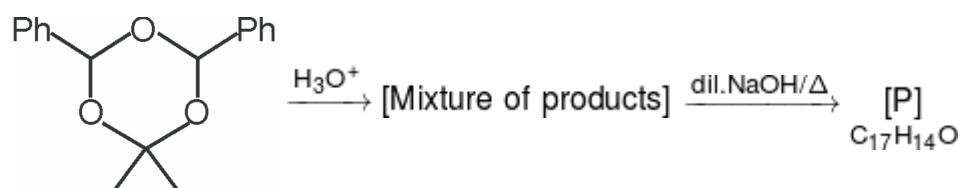


Which statement(s) is/are CORRECT?

- (A) Reduced product of P and Q will be metamers to each other. (Consider LiAlH_4 as a reducing agent)
 (B) By dry distillation of hydrolysed product of P with $\text{Ca}(\text{OH})_2$ gives benzophenone.
 (C) One of the hydrolysis product of Q reacts with $\text{NaNO}_2 + \text{HCl}$ followed by reaction with phenol in mild basic medium gives orange red dye.
 (D) Electrophile involved in the formation of Q is dichlorocarbene.

Space for Rough Work

6. Choose the correct statement(s)



- (A) In mixture, one of the product shows positive iodoform test.
- (B) In mixture, one of the product shows positive Tollen's test and another shows negative Tollen's test.
- (C) [P] on reaction with hydroxyl amine gives oxime that can have 4 stereoisomers.
- (D) Number of C=C bonds in [P] is 10.

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
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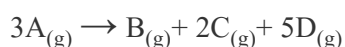
Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

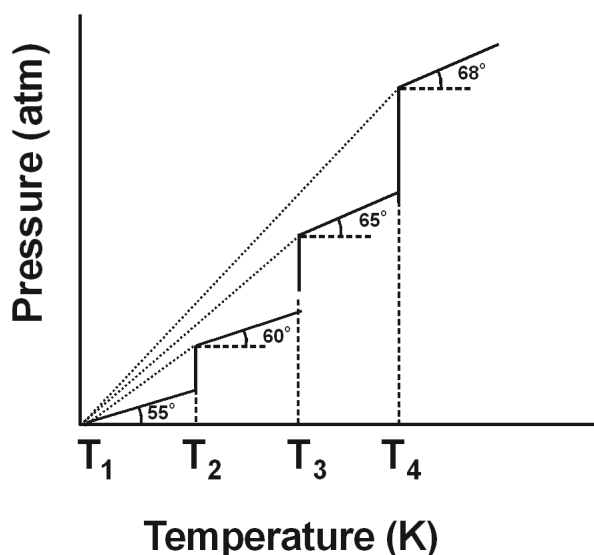
For a gaseous reaction occurring in a rigid vessel



Extent of dissociation depends on temperature.

It is different in different temperature ranges as shown in the shown P vs T plot.

It is constant within a given temperature range



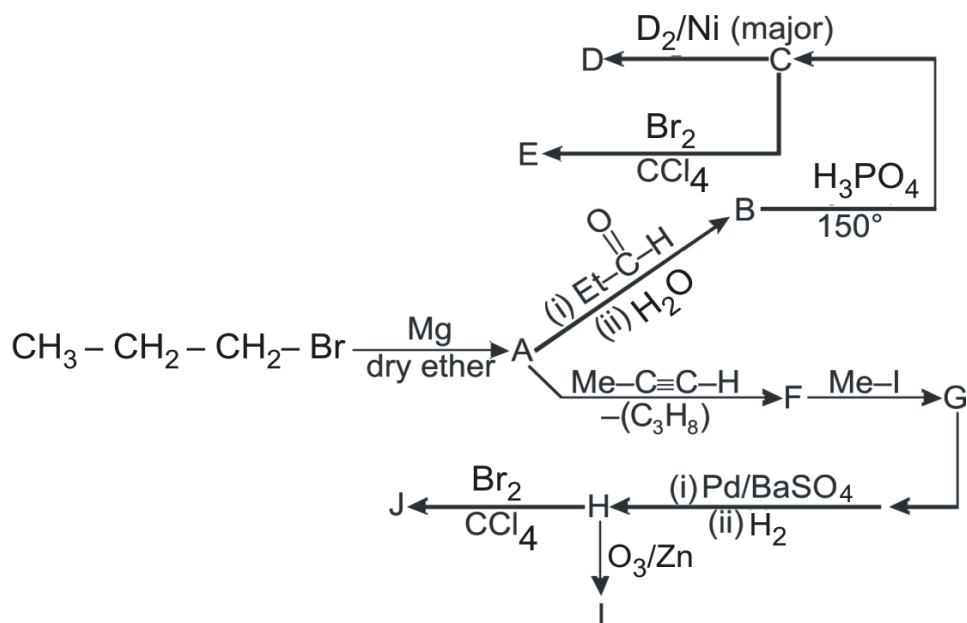
($\tan 55^\circ = 1.43$, $\tan 60^\circ = 1.73$, $\tan 65^\circ = 2.14$, $\tan 68^\circ = 2.47$)

Space for Rough Work

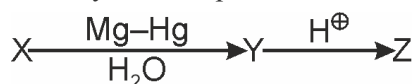
7. If $\alpha_{T_i-T_{i+1}}$ is the degree of dissociation of A then in the temperature range T_i-T_{i+1}
- (A) $\alpha_{T_1-T_2}$ is lowest (B) $\alpha_{T_1-T_2}$ is highest
- (C) $\alpha_{T_4-T_{\infty}} = 1$ (D) $\alpha_{T_4-T_{\infty}} = 0$
8. If initially 1 mole of A is taken in 0.0821 l container then
- (A) $\alpha_{T_2-T_3} = 0.525$ (B) $\alpha_{T_1-T_2} = 0.305$
- (C) $\alpha_{T_1-T_2} = 0.258$ (D) $\alpha_{T_3-T_4} = 0.132$

Space for Rough Work

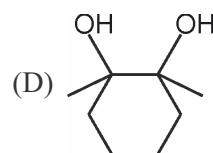
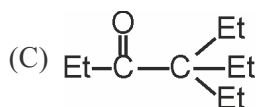
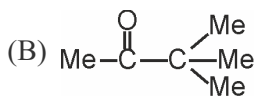
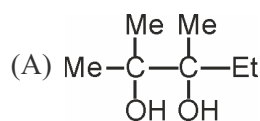
Paragraph for Questions 9 and 10



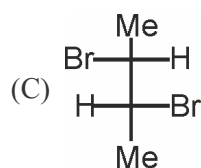
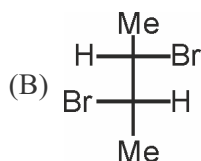
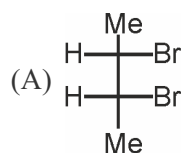
9. Identify "Z" compound



where X is a functional isomer of "W" which is next higher homologue of "I"



10. In the reaction sequence "J" compound is



(D) both (B) and (C)

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

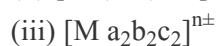
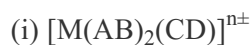
Stereoisomers are the isomers that have the same atoms, the same set of bonds that differ only in the relative orientation of these bonds. Geometrical Isomerism is not exhibited by complexes having tetrahedral geometry. Optical isomerism is possible for both tetrahedral and octahedral complexes, but not for square planar complex. If M represents central metal.

a,b,c,d..... represents monodentate ligands

AA, BB..... represents symmetrical bidentate ligands

AB, BC, CD..... represents unsymmetrical bidentate ligands

1. Consider the following complexes



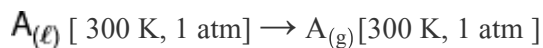
If total number of geometrical isomerism is X and total number of optically active isomer is Y then what is the value of $\left(\frac{X+Y}{2}\right)$.

2. Find total number of possible stereoisomer for the complex compound $[Ma_3bcd]^{n\pm}$

Space for Rough Work

Paragraph for Questions 3 and 4

Consider the following changes for the substance A.



Normal boiling point of A is 400 K.

Enthalpy change for vapourisation of liquid A at 300 K, 1 atm is 30 kJ/mol

$$C_p(A, l) = 50 \text{ J/K} \cdot \text{mol}, C_p(A, g) = 40 \text{ J/K} \cdot \text{mol}$$

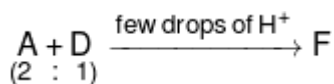
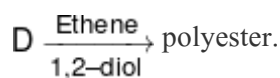
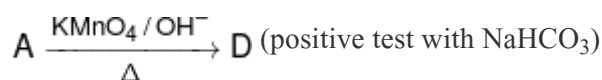
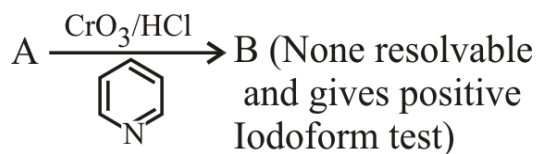
$$[\ln 4 = 1.386, \ln 3 = 1.098]$$

3. Calculate $\Delta S_{\text{vapourisation}}$ (in J/K $\cdot$ mol) of A at 300 K.
4. Calculate enthalpy of vapourisation of A at 400 K in kJ mol⁻¹.

Space for Rough Work

Paragraph for Questions 5 and 6

A is a resolvable organic compound. A has molecular formula $C_{10}H_{14}O$. It does not react with NaOH but with Na metal gives colorless gas. It gives yellow ppt with I_2 and NaOH & does not decolorise Br_2/H_2O .



5. If the molecular weight of 'F' is M then find out the value of $\frac{M}{200}$?
6. How many sp^2 carbon is present in 'D'?

Space for Rough Work

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. [A] is a monomer of natural rubber, on reaction with HBr (1 eq)/CCl₄ at 298 K, it forms major product [B]. The number of allylic hydrogen of [A] is x and that of [B] is y, the value of (y – x)?
8. $\text{FeCr}_2\text{O}_4 + \text{Na}_2\text{CO}_3 + \text{air} \rightarrow (\text{A}) + \text{Fe}_2\text{O}_3 + \text{CO}_2$
 $(\text{A}) + (\text{B}) \rightarrow \text{Na}_2\text{Cr}_2\text{O}_7$
 $\text{Na}_2\text{Cr}_2\text{O}_7 + \text{X} \xrightarrow{\Delta} \text{Cr}_2\text{O}_3$
 $\text{Cr}_2\text{O}_3 + \text{Y} \xrightarrow{\Delta} \text{Cr}$
 Calculate the difference between atomic number of Y and oxidation state of central metal atom of A.
9. A 15% by mass of sucrose solution has undergone partial inversion
 $\text{Sucrose (aq)} + \text{Water} \rightarrow \text{Glucose(aq)} + \text{Fructose (aq)}$
 After 10 hours from the start of reaction the elevation in boiling point of solution is $\frac{103}{200}$ K
 The reaction follows 1st order Kinetics with respect to Sucrose. If 87.5% of inversion of sucrose is complete in 'T' hours then find value of $\left(\frac{T}{2.5}\right)$?
 $K_b(\text{water}) = 0.57 \text{ kg K mol}^{-1}$

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. A box contains n coins of which at least one is biased. Let E_k denote the event that exactly k out of the n coins are biased. Let $P(E_k)$ is proportional to k for $1 \leq k \leq n$. If A denotes the event that a coin selected out of n coins at random is biased, then :-

(A) $\lim_{n \rightarrow \infty} P(A) = \frac{1}{3}$

(B) $\lim_{n \rightarrow \infty} P(A) = \frac{2}{3}$

(C) $P\left(\frac{E_1}{A}\right) = \frac{3}{n(n+1)(2n+1)}$

(D) $P\left(\frac{E_1}{A}\right) = \frac{6}{n(n+1)(2n+1)}$

Space for Rough Work

2. Let DABC is a tetrahedron. $A = (2, 0, 0)$, $B = (0, 4, 0)$ and $CD = \sqrt{14}$. Edge CD lies on the line $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-3}{3}$. If locus of centroid of tetrahedron is $\frac{x-\frac{3}{2}}{1} = \frac{y-y_1}{a} = \frac{z-z_1}{b}$ then which of the following is(are) correct ?
- (A) $a + b = 5$ (B) $y_1 + z_1 = 6$ (C) $y_1 - z_1 = 1$ (D) $a + b + y_1 = 8$
3. During the weekends, Mr. A delivers news paper in the complex plane. On Saturday, he being at z and delivers news paper to houses located at $z^3, z^5, z^7, \dots, z^{2017}$ in that order. On Sunday, he being at 1 and delivers news paper to houses located at $z^2, z^4, z^6, \dots, z^{2016}$ in that order. Mr. A travels in a straight line between two houses. The distance he must travel from his starting point to the last house is $\sqrt{2016}$ on each of the two days, then :-
- (A) $|z| = 1$ (B) $|z| = \frac{1}{2}$ (C) $\operatorname{Re}(z^2) = \frac{1008}{1009}$ (D) $\operatorname{Re}(z^2) = \frac{1007}{1008}$
4. Let $f(x)$ be a cubic polynomial with leading coefficient unity such that $f(a) = b$ and $f'(a) = f''(a) = 0$. Suppose $g(x) = f(x) - f(a) + (a-x)f'(x) + 3(x-a)^2$ for which conclusion of Rolle's theorem in $[a, b]$ holds for $x = 2 \in (a, b)$, then
- (A) $f''(2) = 6$ (B) $a + 2b = 6$
- (C) $\int_0^2 f(x) dx = 2b$ (D) number of roots of $f(f(x)) = 0$ is 2

Space for Rough Work

5. Let $f(x) = \lim_{\alpha \rightarrow 0} \frac{1}{\alpha^4} \int_0^\alpha \frac{(e^{x+t} - e^x) \sin t \cdot \ln(1+t) \cdot (2017t^3 + 4)}{(2018t^3 + 1)} dt$ and $g(x)$ be a real valued

function and satisfying $g(x) = x^2 + \int_{-1}^1 (x+y) g(y) dy$, then :-

[Where $[z]$ denotes greatest integer function less than or equal to z .]

- (A) $[f(1)] = 2$
 (B) $[g(1)] = -3$
 (C) number of solutions of $f(x) = g(x+1) + \frac{7}{3}$ is 1
 (D) number of solutions of $f(x) = g(x+1) + \frac{7}{3}$ is 2
6. If a continuous function $f(x)$ satisfies the relation $\int_0^x t f(x-t) dt = \int_0^x f(t) dt + \sin x + \cos x - x - 1; \forall x$, then which of the following does not holds good ?

- (A) $f(0) = 1$ (B) $f'(0) = 0$ (C) $f''(0) = 2$ (D) $\int_0^\pi f(x) dx = e^\pi$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

Let π_1 denotes the equation of the plane to which the vector $(1, 1, 0)$ is normal and which contains the line L whose equation is $\vec{r} = \hat{i} + \hat{j} + \hat{k} + \lambda (\hat{i} - \hat{j} - \hat{k})$. π_2 denotes the equation of the plane containing the line L and a point with position vector $(0, 1, 0)$.

7. Vector of magnitude $\sqrt{6}$ along the line of intersection of planes π_1 and π_2 and perpendicular to normal vector of plane π_1 , is
- (A) $\pm\sqrt{2}(-1, 1, 1)$ (B) $\pm\sqrt{2}(1, -1, 1)$
 (C) $\pm\sqrt{2}(1, 1, -1)$ (D) $\pm\sqrt{2}(1, 1, 1)$
8. The acute angle between plane π_1 and plane π_2 , is
- (A) $\tan^{-1} 1 - \tan^{-1} (\sqrt{2} - 1)$ (B) $\tan^{-1} 1 - \tan^{-1} (2 - \sqrt{3})$
 (C) $\tan^{-1} 1 - \tan^{-1} (\sqrt{2} + 1)$ (D) $\tan^{-1} 1 - \tan^{-1} (2 + \sqrt{3})$

Space for Rough Work

Paragraph for Questions 9 and 10

If $f(x)$ be a polynomial function satisfying the relation $f(x) \cdot f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$ and $f(10) = 101$, then

9. The value of $\int_{f(0)}^{f(1)} \frac{dx}{f(x) + 3}$ is

- (A) $\frac{1}{2} \tan^{-1} 3$ (B) $\frac{1}{2} \cot^{-1}(3)$ (C) $\frac{1}{2} \tan^{-1} \left(\frac{2}{\sqrt{3}} \right)$ (D) $\frac{1}{2} \left(\frac{\pi}{2} - \cot^{-1} 2 \right)$

10. Two real numbers x and y are selected out at random such that $0 \leq x \leq 5$, $0 \leq y \leq 5$. Then the probability that $y \geq f(x)$ is

- (A) $\frac{1}{3} \left(\frac{f(4) - f(0)}{f(5) - f(0)} \right)$ (B) $\frac{f(4) - f(0)}{f(5) - f(0)}$ (C) $\frac{f(3) + f(0)}{f(5) + f(0)}$ (D) $\frac{1}{3} \left(\frac{f(3) - f(0)}{f(4) + f(1)} \right)$

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

If a chord of the parabola $y^2 = 4x$ subtend a right angle at the centre of the circle $x^2 + y^2 = 16$ and the chord passes through a fixed point P and from point P three normal are drawn to intersect the parabola at A, B, C. Then

1. If area of ΔABC is Δ , then the value of $[\Delta]$ is (where $[x]$ is greatest integer less than or equal to x)
2. Out of the 3 normal chord, if the length of normal chord which is minimum is ℓ then $\ell\sqrt{3}$ is

Space for Rough Work

Paragraph for Questions 3 and 4

Let f be a function defined by $f(x) = \int_0^{\pi/2} |x - 2 \sin t| \cos t \, dt$ & $g : \mathbb{R} \rightarrow \mathbb{R}$ be a derivable function

which satisfies $g(x) = 1 + \int_0^x g(t) \, dt$ and $g'(0) = 1$

3. If $f(|x|) = k$ has 4 distinct solutions, where $k \in (a, b)$, then the value of $a + b$ is :-
4. Number of distinct solutions of $f(x) = g(\ln 10) + g'(\ln 10) + g''(\ln 10)$ is equal to :-

Paragraph for Questions 5 and 6

Let $X = \{1, 2, 3\}$. A number is selected from set X with replacement and this process is done 1100 times. For each i , $1 \leq i \leq 3$. Let $f(i)$ represent the number of times the number i is selected. Clearly $f(1) + f(2) + f(3) = 1100$. Also, let S denote the total sum of 1100 numbers selected. If $S^3 = 162 \cdot f(1) \cdot f(2) \cdot f(3)$.

5. The value of harmonic mean of $f(1)$, $f(2)$ and $f(3)$ is
6. There are $f(1)$ number of red balls, $f(2)$ number of blue balls and $f(3)$ number of white balls in a bag. Two persons A & B in order draws a ball from the bag & put it back after noting its colour. Person who draws a white ball first wins the game. If A starts the game, then probability of winning of B is :-

Space for Rough Work

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. Let A be a point inside a regular polygon of 10 sides. Let $p_1, p_2, \dots, p_{10}$ be the distances of A from the sides of the polygon. If each side is of length 2, then find the minimum possible integral value of $\frac{1}{p_1} + \frac{1}{p_2} + \dots + \frac{1}{p_{10}}$
8. If radius of a circle whose centre is (x_0, y_0) touches a parabola $y^2 = 4x$, a straight line $x - y + 1 = 0$ and the y-axis is $\sqrt{p} \tan \frac{\pi}{8}$; where $x_0 - y_0 + 1 < 0$, then find the value of p :-
9. If a and b be coprime numbers and $\lim_{x \rightarrow \frac{\pi}{2}^-} (1 + \tan x) \left((1 + \tan x) \log \left(\frac{1 + \tan x}{2 + \tan x} \right) + 1 \right) = \frac{a}{b}$, then find the minimum value of $(a^2 + b^2)$.

Space for Rough Work

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work

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